$S°(298.15\ K) = 187.86 \pm 0.5\ J \cdot K^{-1} \cdot mol^{-1}$

$\Delta_f H°(0\ K) = 259.5 \pm 20\ kJ \cdot mol^{-1}$

$\Delta_f H°(298.15\ K) = 259.4 \pm 20\ kJ \cdot mol^{-1}$

Electronic Level and Quantum Weight		
State	ϵ_i, cm^{-1}	g_i
$^1\Sigma^+$	0	1

$\omega_e = 1682.56\ cm^{-1}$ $\omega_e x_e = 29.09\ cm^{-1}$ $\sigma = 1$

$B_e = 6.3907\ cm^{-1}$ $\alpha_e = 0.1858\ cm^{-1}$ $r_e = 1.6477\ Å$

Enthalpy of Formation

Gaydon[1] extrapolated the available $X^1\Sigma$ vibrational levels, also considered predissociation in the A^1 state and concluded that the best value of $D_0°$ was 2.9 ± 0.2 ev. The linear Birge–Sponer extrapolation gave a value of 3.05 ev while the predissociation limit was 3.07 ev. The value recommended by Gaydon[1] is adopted here and leads to a value of $\Delta_f H°$(AlH, g, 298.15 K) = 62.16 ± 0 5.0 $kcal \cdot mol^{-1}$, using JANAF auxiliary data.

Heat Capacity and Entropy

Zeeman and Ritter[2] examined the UV band system of AlH under high resolution and reported the molecular constants adopted above.

References

[1] A. G. Gaydon, 'Dissociation Energies', 2nd ed., Chapman and Hall, Ltd., London, 261 pp. (1953).
[2] P. B. Zeeman and E. G. Ritter, Can. J. Phys. 32, 555 (1954).

Enthalpy Reference Temperature = T_r = 298.15 K Standard State Pressure = $p°$ = 0.1 MPa

	$J \cdot K^{-1} mol^{-1}$			$kJ \cdot mol^{-1}$			
T/K	$C_p°$	$S°$	$-[G°-H°(T_r)]/T$	$H°-H°(T_r)$	$\Delta_f H°$	$\Delta_f G°$	log K_f
0	0.	0.	INFINITE	−8.669	259.512	259.512	INFINITE
100	29.124	156 000	213.828	−5.783	260.415	250.550	−130.874
200	29.154	176.194	190.540	−2.869	260.180	240.711	−62.867
250	29.220	182.705	188.346	−1.410	259.827	235.882	−49.285
298.15	29.373	187.864	187.864	0	259.408	231.308	−40.524
300	29.381	188.045	187.864	0.054	259.391	231.133	−40.244
350	29.658	192.594	188.223	1.530	258.907	226.462	−33.798
400	30.041	196.578	189.023	3.022	258.397	221.861	−28.972
450	30.498	200.142	190.064	4.535	257.877	217.325	−25.226
500	30.996	203.381	191.236	6.072	257.354	212.848	−22.236
600	32.008	209.122	193.751	9.223	256.304	204.045	−17.764
700	32.942	214.128	196.312	12.471	255.236	195.419	−14.582
800	33 754	218.581	198.822	15.807	254.115	186.949	−12.207
900	34.441	222.597	201.244	19.218	252.897	178.626	−10.367
1000	35.018	226.257	203.565	22.692	240.954	171.204	−8.943
1100	35.505	229.618	205.783	26.218	239.786	164.286	−7.801
1200	35.917	232.725	207.900	29.790	238.643	157.473	−6.855
1300	36.271	235.615	209.922	33.400	237.518	150.755	−6.057
1400	36.577	238.314	211.855	37.043	236.403	144.123	−5.377
1500	36.844	240.847	213.704	40.714	235.296	137.570	−4.791
1600	37.081	243.232	215.476	44.410	234.191	131.091	−4.280
1700	37.293	245.487	217.176	48.129	233.089	124.681	−3.831
1800	37.484	247.624	218.808	51.868	231.985	118.336	−3.434
1900	37.658	249.655	220.379	55.626	230.881	112.052	−3.081
2000	37.818	251.591	221.891	59.399	229.775	105.826	−2.764
2100	37.966	253.440	223.350	63.189	228.667	99.656	−2.479
2200	38.104	255.209	224.758	66.992	227.556	93.539	−2.221
2300	38.234	256.906	226.119	70.809	226.442	87.472	−1.987
2400	38.356	258.536	227.436	74.639	225.326	81.454	−1.773
2500	38.473	260.104	228.712	78.480	224.207	75.482	−1.577
2600	38.584	261 615	229.948	82.333	223.086	69.555	−1.397
2700	38.691	263.073	231.148	86.197	221.963	63.671	−1.232
2800	38.793	264.482	232.314	90.071	−73.064	58 797	−1.097
2900	38.893	265 845	233.447	93.955	−73.096	63.507	−1.144
3000	38.989	267.165	234.549	97.850	−73.130	68.218	−1.188
3100	39.082	268.445	235.622	101.753	−73.166	72.930	−1.229
3200	39.174	269.688	236.667	105.666	−73.205	77.644	−1.267
3300	39.263	270 894	237.686	109.588	−73.245	82.358	−1.304
3400	39.350	272.068	238.680	113.518	−73.288	87 074	−1.338
3500	39.435	273.210	239.650	117.458	−73.333	91.791	−1.370
3600	39.520	274.322	240.598	121.406	−73.380	96.510	−1.400
3700	39.602	275.406	241.524	125.362	−73.430	101.230	−1.429
3800	39.684	276.463	242.430	129.326	−73.483	105.951	−1.456
3900	39.764	277.495	243.316	133.298	−73.540	110 674	−1.482
4000	39.844	278.503	244.183	137.279	−73.600	115.398	−1.507
4100	39.922	279.487	245.032	141.267	−73.664	120.124	−1.530
4200	40.000	280.450	245 864	145.263	−73.732	124.851	−1.553
4300	40.077	281.393	246.679	149.267	−73.805	129.580	−1.574
4400	40.154	282.315	247.479	153.279	−73.883	134.311	−1.594
4500	40.229	283.218	248.263	157.298	−73.965	139.043	−1.614
4600	40.305	284.103	249.032	161.324	−74.055	143.778	−1.633
4700	40.379	284.971	249.788	165.359	−74.152	148.514	−1.651
4800	40.454	285.821	250.530	169.400	−74.256	153.253	−1.668
4900	40.527	286.656	251.259	173.449	−74.367	157.994	−1.684
5000	40 601	287.476	251.975	177.506	−74.488	162.737	−1.700
5100	40.674	288.281	252.679	181.570	−74.618	167.483	−1.715
5200	40.746	289.071	253.371	185.641	−74.758	172.231	−1.730
5300	40.819	289.848	254.052	189.719	−74.909	176.982	−1.744
5400	40.891	290.612	254.722	193.804	−75.071	181.737	−1.758
5500	40.963	291.363	255.381	197.897	−75.246	186.494	−1.771
5600	41.034	292 101	256.030	201.997	−75.433	191.254	−1.784
5700	41.106	292.828	256.670	206.104	−75.634	196.018	−1.796
5800	41.177	293.544	257.299	210.218	−75.846	200.786	−1.808
5900	41.247	294.248	257.920	214.339	−76.076	205.557	−1.820
6000	41.318	294.942	258.531	218.467	−76.300	210.331	−1.831

PREVIOUS: June 1963 (1 atm) CURRENT June 1963 (1 bar)